



Computer Networks

计算机网络 (CST2398)

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03

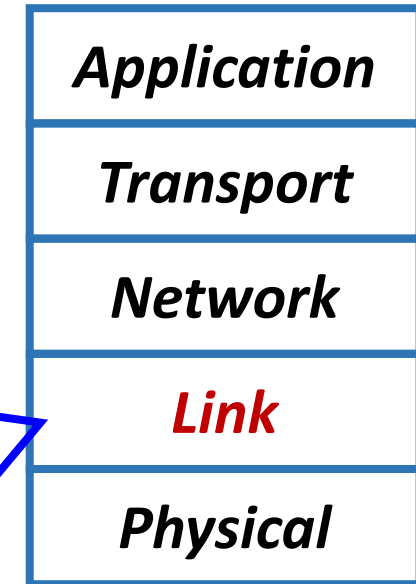
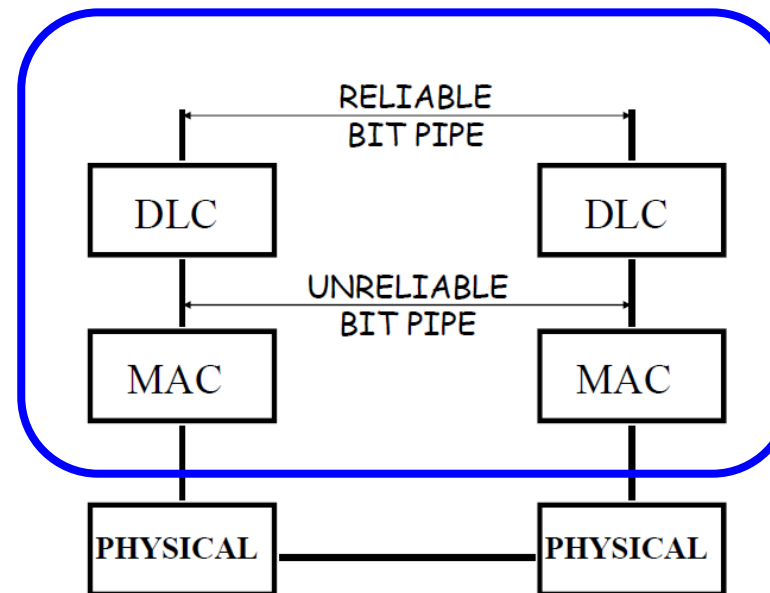
Medium Access Control (MAC)

Contents of this Chapter

- Functions of **Medium Access**

- Control (MAC):**

- **Allocation** of a single broadcasting channel
 - Also referred to as the “multiple access” protocols.
 - Fixed: “**XDMA**” : poor performance in variable environments.
 - **Dynamic channel allocation**



MAC: Channel Allocation of a Single Broadcasting Channel

Who can transmit at when?

Multiple Access Links

- **Point-to-Point (P2P) links**
 - PPP for dial-up access
 - point-to-point link between Ethernet switch and host
- **Broadcast (a shared wire or medium)**
 - old-fashioned Ethernet (Bus 总线 type Ethernet)
 - 802.11 wireless LAN



shared wire (e.g.,
cabled Ethernet)



shared RF
(e.g., 802.11 WiFi)



shared RF
(satellite)

Multiple Access Protocols

- *The ideal case for a channel of capacity R (max data rate R)*

What we want

- When *one node* wants to transmit, it can send at *rate R* .
- When *M nodes* want to transmit, each can send at an *average rate R/M* .
- Fully *decentralized*/ad hoc
 - No special node to coordinate
 - No synchronization of clocks/slots
- *Simple*

- *Multiple Access Protocols*

- ALOHA

Random Access MAC protocols

- *Carrier Sensing Multiple Access (CSMA)*

- Taking Turns Protocols

- *Collision*-free Protocols

- Limited-Contention Protocols

- *Ethernet Protocols*

- *Wireless LAN Protocols*

Reality



3.1

Random Access MAC

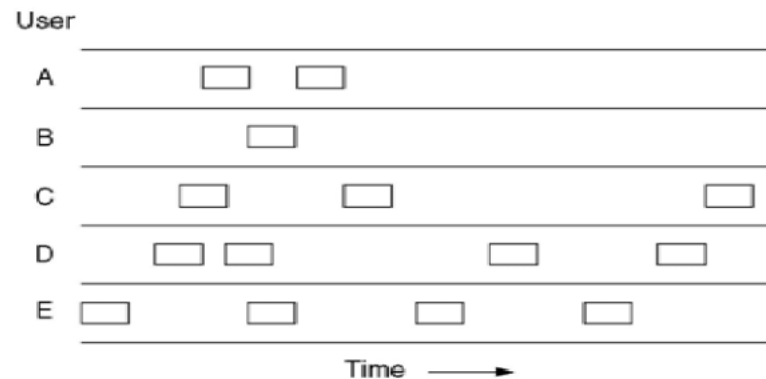
Random Access Protocols

- **When nodes have data (frame) to transmit**
 - They transmit at data rate R
 - No coordination a priori (so there will be possible **冲突** *collision* & **竞争** *contention*!)
- **Collision**: When 2 or more nodes are transmitting simultaneously
- **Contention System**: multiple users share a common channel in a way that leads to conflicts.
- **Key questions/ what random access protocols specify:**
 1. How to **detect** collisions?
 2. How to **recover** from collisions?

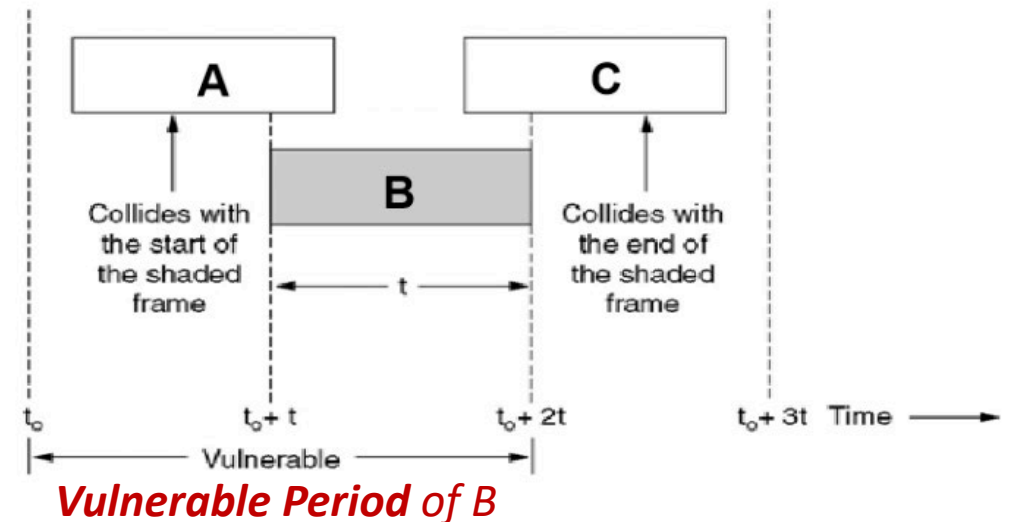
Pure ALOHA

- Users transmit at will. -> Collisions happen.
- **Detecting** collisions: **listening** to the broadcast channel.
- **Recovering**: **wait** for a **random** period of time **and retry**.

Its performance
when users are
few/many?



- **How does a collision happen?**



Throughput of Pure ALOHA

• Assumptions:

- Frames are of fixed lengths, the amount of time (**frame time**) needed to transmit a frame is t .
- Worst-case Scenario: There are **infinite users** in the system.
- **New frames** are generated according to a **Poisson Process** 泊松过程 with mean N (frames/frame time).
 - **$0 \leq N < 1$** : Otherwise the channel can not handle and all frames will be damaged.

Simple, but
not good

• Analysis of its throughput:

- **Offered Load G** : (considering **retransmissions**) all the frames are generated with mean G (frames/frame time).
 - Low load: $N \simeq 0$ (few collisions) - $\rightarrow G \simeq N$
 - High load: $N \simeq 1$ (many collisions) - $\rightarrow G \gg N$
 - It's a Poisson process as well. Probability that k frames are generated during a frame time is $P(k) = G^k e^{-G} / k!$
- Probability of a successful transmission: no other nodes transmit during a frame's **vulnerable period**:
$$P_S = P(k = 0) \cdot P(k = 0) = e^{-2G}$$
- **Throughput S** : average # of successful transmissions per frame time. $S = G P_S = G e^{-2G}$

$$S \leq 1/2e$$

Slotted ALOHA

- **Time is cut into slots:** when a node wants to transmit a frame, it has to transmit at the beginning of the next time slot.

- Nodes are synchronized.
- Collision happens when two or more nodes transmitted during a time slot.

- Probability of no collision (successful transmission): $P_S = e^{-G}$

Why?

Binomial
二项分布

- Probability of collision: $1 - e^{-G}$

- **Advantage:** Vulnerable period is now 1! $\rightarrow$ Throughput $S = GP_S = Ge^{-G}$

- Number of attempts/transmissions until success (r.v.) X :

$$P(X = k) = (1 - P_S)^{k-1} P_S = (1 - e^{-G})^k e^{-G}$$

So its expectation 期望 $E(X) = \sum_k kP(X = k) = \frac{1}{P_S} = e^G$

Exponential

When G changes a little, $E(X)$ changes a lot (a lot of retransmissions).

Throughput v.s. Offered Load

- **The max throughput of slotted ALOHA and pure ALOHA.**

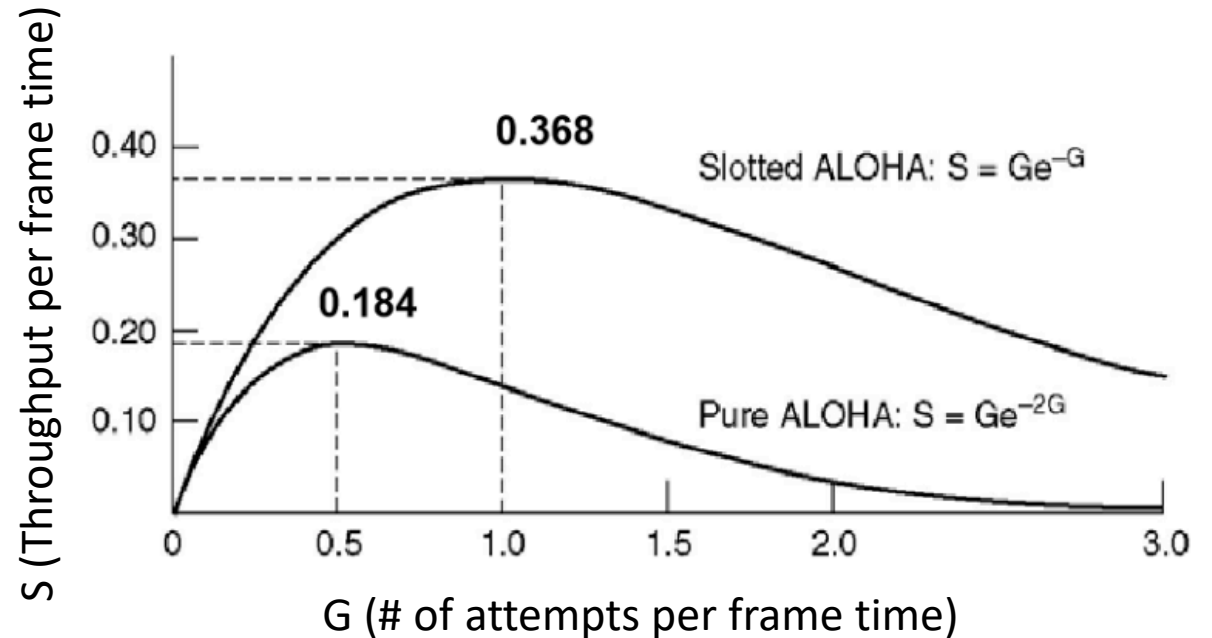
- Pure ALOHA

$$S = GP_S = Ge^{-2G}$$

- Slotted ALOHA

$$S = GP_S = Ge^{-G}$$

Why? vulnerable
time: 2 → 1



- **Each user transmits at will → Throughput (or channel utilization) is low**

Carrier Sensing Multiple Access (CSMA)

- **How to improve channel utilization? -> Check before transmit!**

- **Application scenario:** LANs - propagation delay is small.

- Nodes can **detect** an idle period and **then transmit**.
- Nodes are **not allowed to transmit** if the channel is **busy**.



- **Carrier Sensing Multiple Access (CSMA)** 载波侦听多址接入

$p=1$

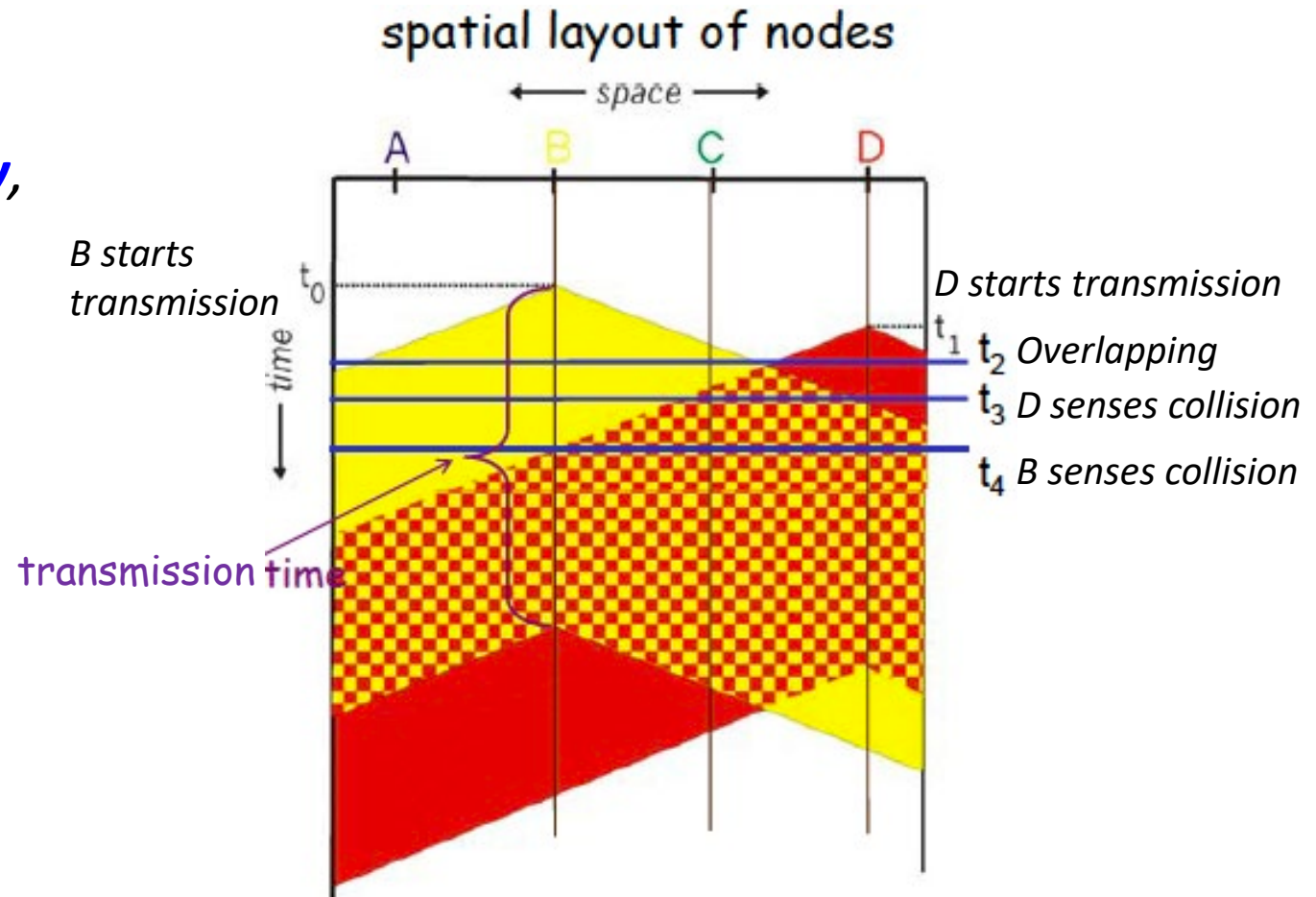
- **1-Persistent CSMA:** if the node **detects** the channel **idle**, it **transmits** a frame. Otherwise, the node waits (**keeps listening**) until the channel becomes idle (so it can send immediately).
- **Non-persistent CSMA:** the node does not listen continuously, but checks the channel again after a random period of time.
- **p -Persistent CSMA:** if the node detects the channel idle, it transmits a frame **with probability p** (and defer/wait with probability $1-p$).

Can this prevent collision?

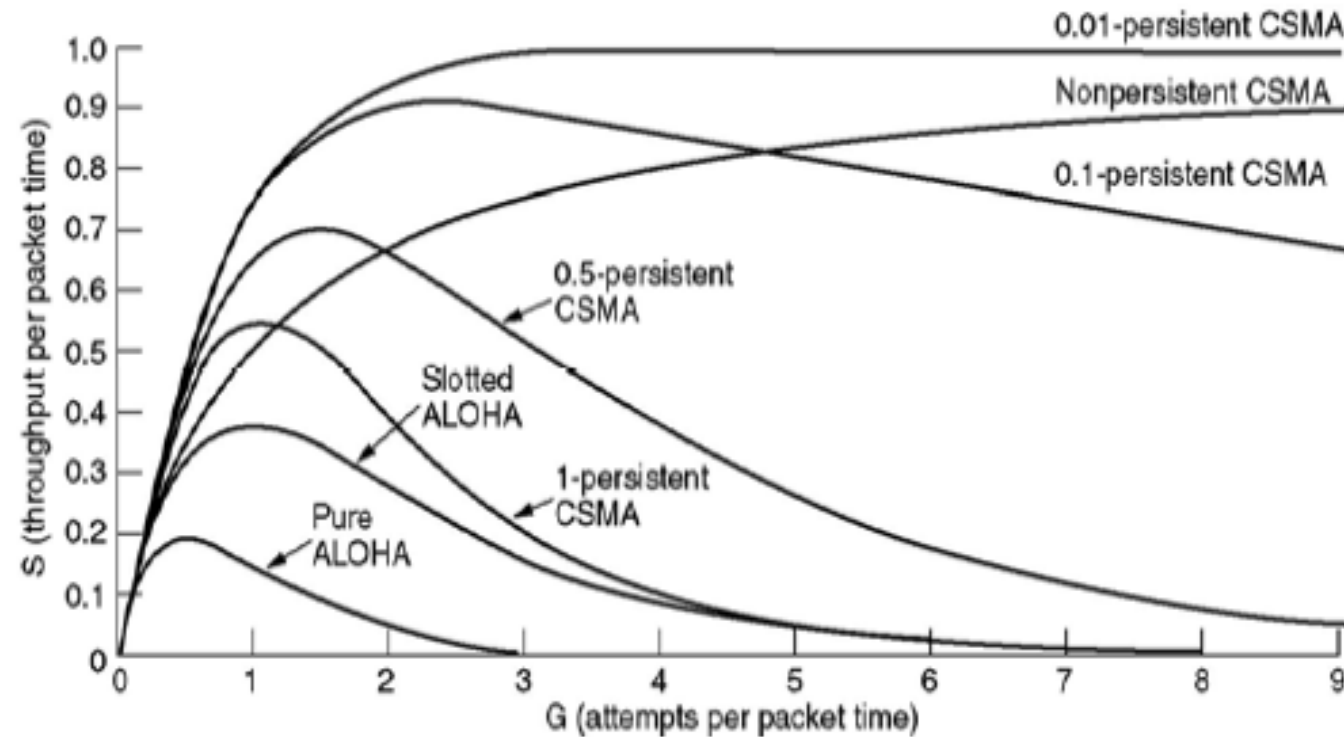
No, collisions may still happen

- **Why? propagation delay**
 - **distance** and **propagation delay**,
 - Two nodes may not hear each other's transmissions.
 - Collision happens ->

The entire frame transmission time is wasted. ☹️



Throughput Comparison



Consequences?
Long waiting
time!

CSMA is not enough!

Comparison of the channel utilization versus load
for various random access protocols.

CSMA/CD (with Collision Detection)

- **CSMA/CD**: carrier sensing and deferral as in CSMA

- If collision is detected -> abort transmission immediately.
- Analogy: a polite conversation - stop talking if others are talking.

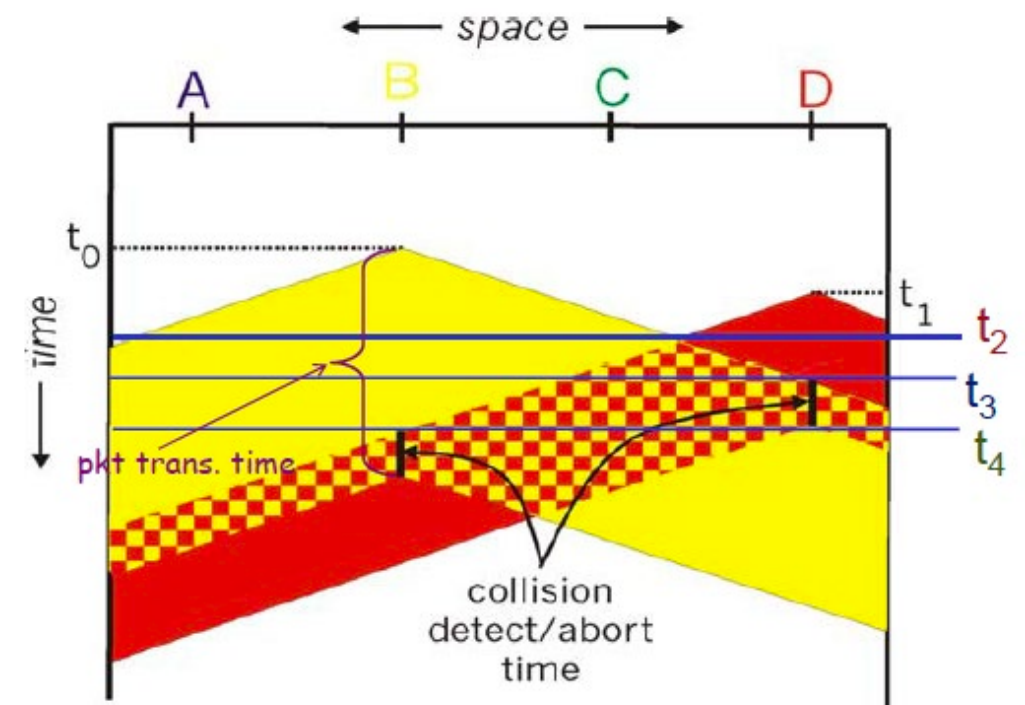
- **Collision Detection** - How:

Easy

- **Wired LAN**: Measure the signal strength and compare with what is transmitted.

Hard

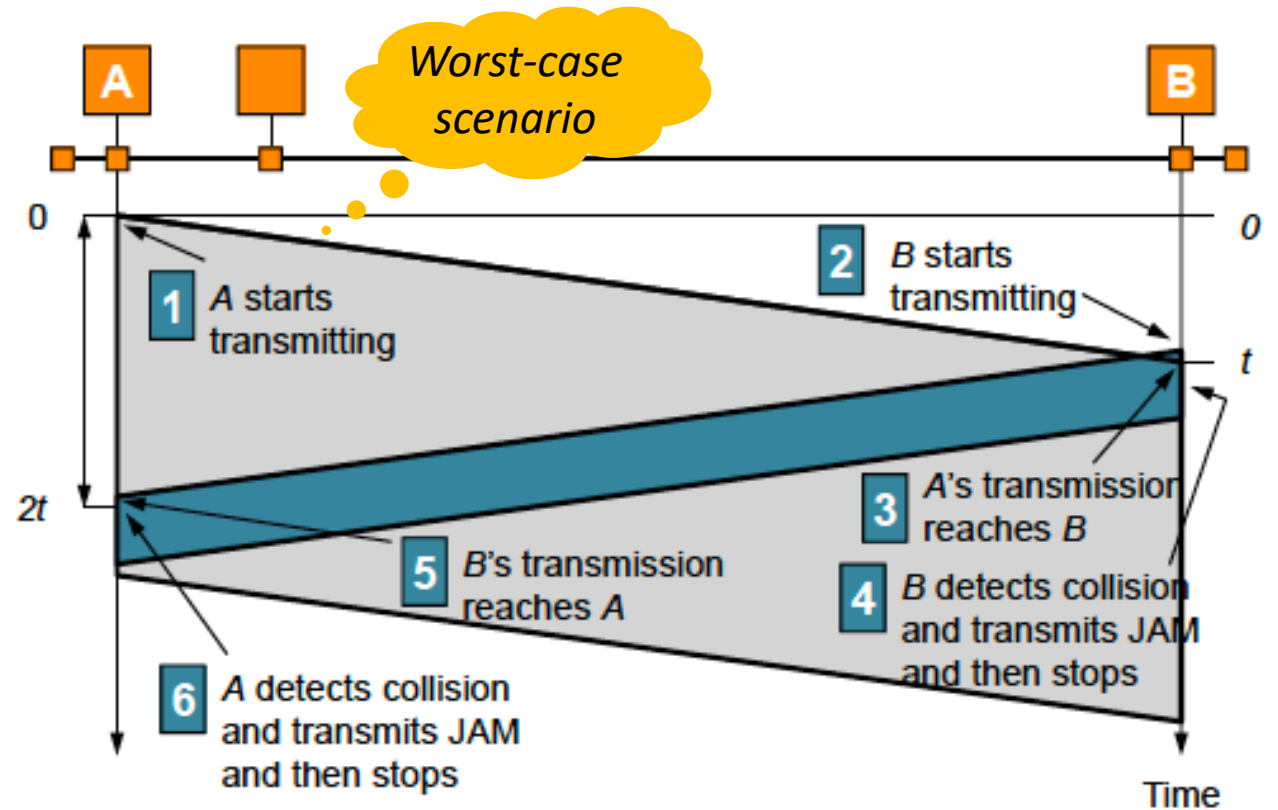
- **Wireless LAN**: Difficult to do so, because the receiver is shut off while transmitting.



Design logic:
Do not waste time

Collision Detection Time

- *How long can two nodes realize there is a collision?*
 - Min: propagation delay t
 - Max: RTT, about $2t$
- **CSMA/CD $\simeq$ slotted ALOHA with slot of length $2t$**
 - Can send more data
 - Good for systems with short propagation delays

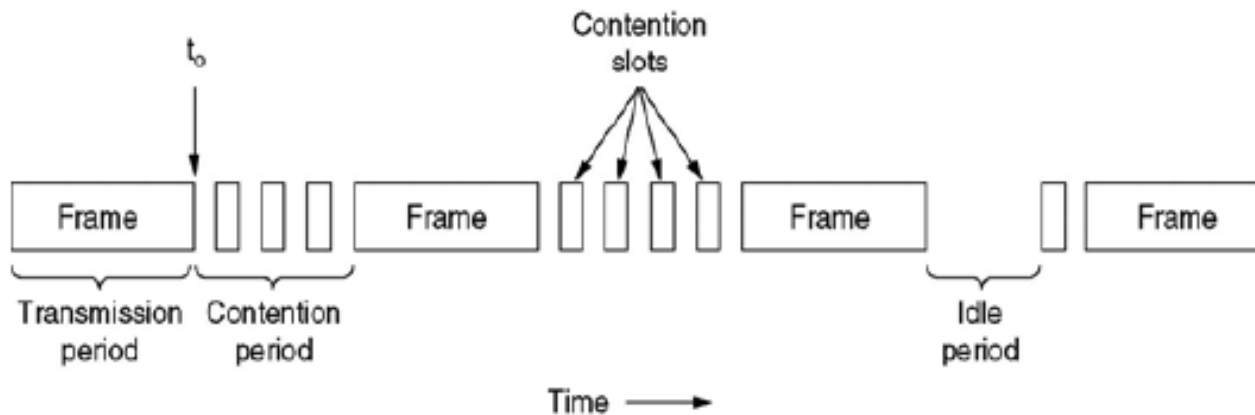


Length of contention slot = $2t$

E.g. max collision detection time

CSMA/CD Operations

- **Three states of CSMA/CD systems**
- **For a node to transmit a frame:**



- **Wait** until the channel is **idle**.
- When the channel is idle, **transmit and listen** while transmitting.
- In case of collision, stop transmission, and **wait for a random period of time (defer)** to sense the channel again.

- **Contention**
- **Transmission**
- **Idle**

Comparison of MAC Protocols

- (Static) **Channel partitioning** MAC protocols
 - **high load**: channel is efficiently and fairly shared -> high efficiency
 - **low load**: delay in channel access, $1/N$ bandwidth allocated even if only 1 active node -> low efficiency
- **Random access** MAC protocols: ALOHA, CSMA/CD, CSMA/CA...
 - **high load**: collisions -> more overhead -> low efficiency
 - **low load**: single node can fully utilize channel -> high efficiency
- **Contention-free protocols**
 - look for best of both worlds!

For wireless,
will explain
later in this
chapter



3.2

Contention-Free MAC

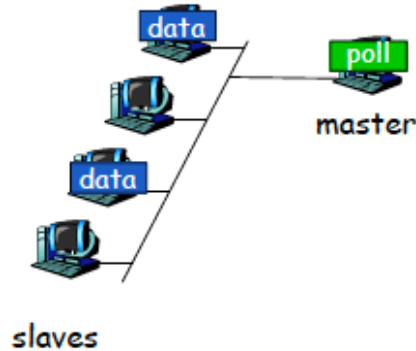
“Taking Turns” MAC protocols

• Polling 轮询

- Master node invites slave nodes to transmit in turn
- Typically used when slaves are “dumb”

• Problems:

- overhead
- latency
- single point of failure (master) -> system fails



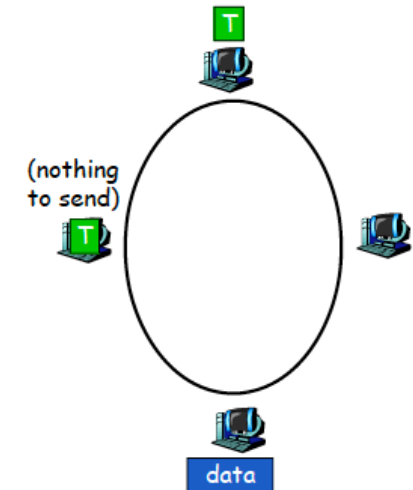
• Token Passing

- Control **token** (令牌) is passed from one node to the next one sequentially.

• Token message

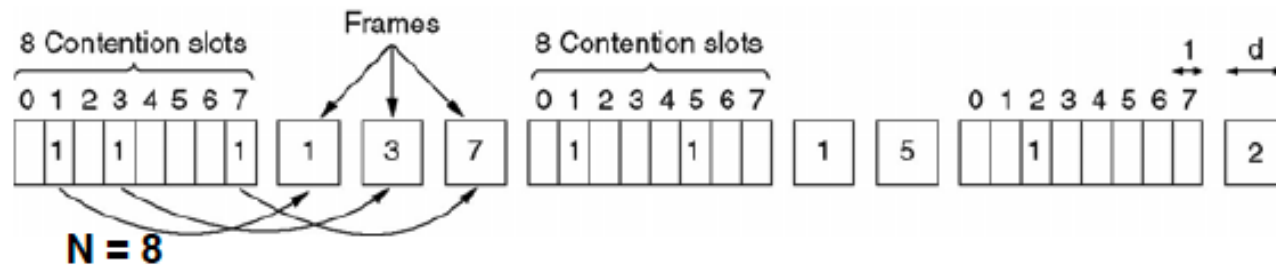
• Problems:

- overhead
- latency
- single point of failure (token) -> system fails



Bit-map and Binary Countdown Protocols

- **Bit-map: Reservation-based**



- *Efficiency: consider N users, each frame is d bits long:*

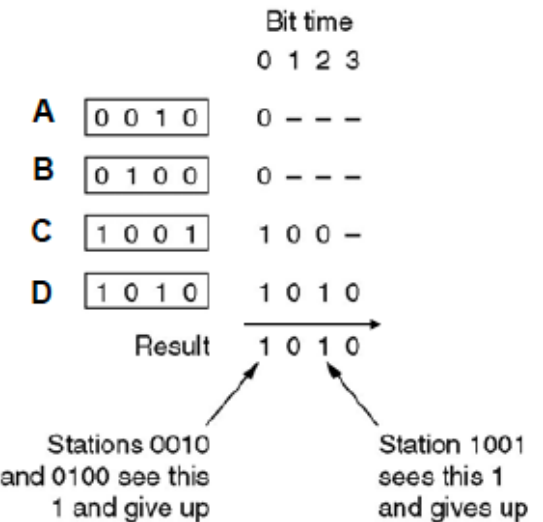
- *Low load: low efficiency*

$$\frac{kd}{kd+N} \xrightarrow{k \text{ small}} \frac{d}{d+N}$$

- *High load: high efficiency*

$$\frac{kd}{kd+N} \xrightarrow{k \rightarrow N} \frac{d}{d+1}$$

- **Binary Countdown: Prioritized, scales well**



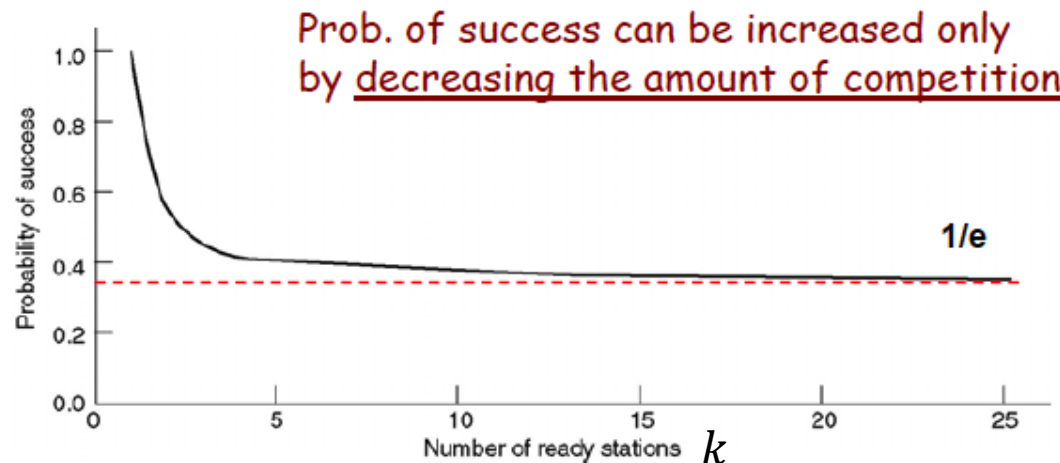
- **Operations**

- *Each node has a binary address of the same length.*
- *Bits in each address position from different stations are OR'ed together.*

- *Efficiency: $d/(d+\log_2 N)$*

Limited-Contention Protocols

- **Trade-offs as load varies**
 - Low load: contention is preferred -> low delay
 - High load: contention-free is preferred -> high channel utilization
- **Limited-contention protocols: a combination**



$$\alpha(p) = kp(1-p)^{k-1}$$
$$\max_p \alpha(p) = \left(1 - \frac{1}{k}\right)^{k-1}$$
$$\lim_{k \rightarrow \infty} \left(1 - \frac{1}{k}\right)^{k-1} = \frac{1}{e}$$

Acquisition probability for a symmetric contention channel.



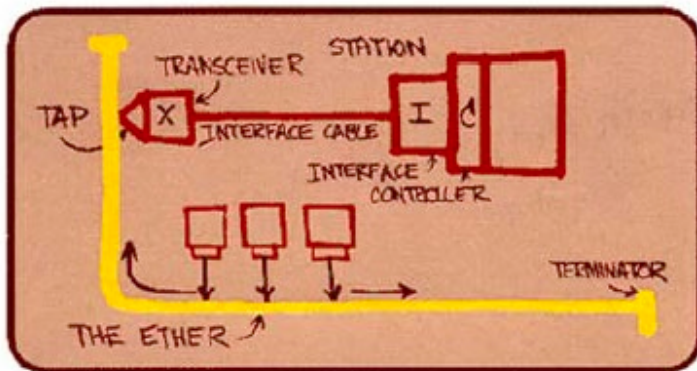
3.3

Ethernet

Ethernet

- **The “dominant” wired LAN technology**

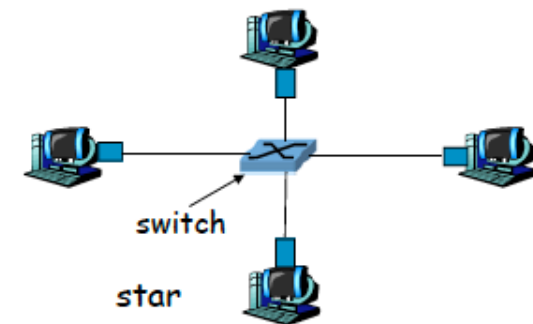
- cheap - NIC (network interface card) ~ ¥ 100
- the first widely used LAN technology
- simpler and cheaper than token LAN and ATM
- improving speed (data rate): 10 Mbps - 10 Gbps
- backward compatible



Metcalfe's Ethernet sketch

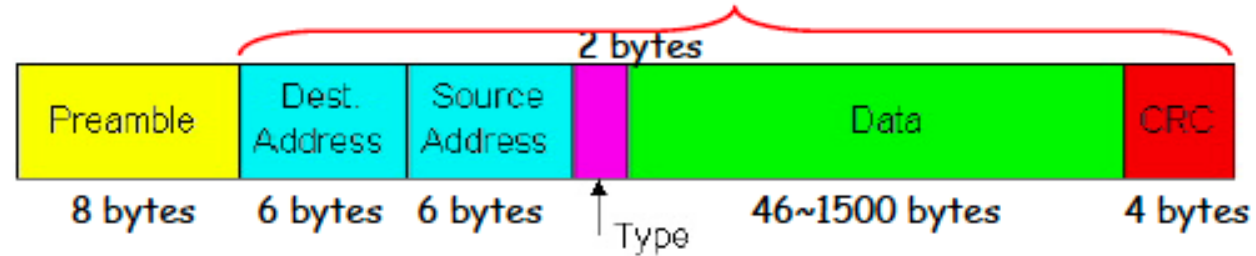
- **Topology**

- Bus 总线 - mid 90's
- Star - popular now



Ethernet Frame Structure

- **Encapsulation: IP datagram -> Ethernet frame**



- **Preamble**: delineate frames, synchronize receiver/sender clock rates
- **Address**: 6 bytes MAC address (format AA-AA-AA-AA-AA-AA)
- **Type**: higher-layer protocol type (IP, Apple Talk, etc.)
- **Data**: 46 - 1500 bytes, **max transmission unit (MTU)=1500 bytes**
- **CRC**: checked at the receiver, a frame is dropped when an error is detected.

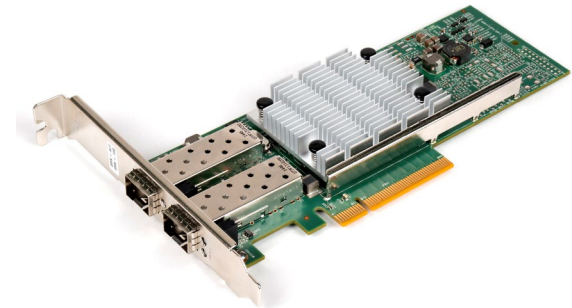
What's the max and min frame length?

MAC Addressing

- *MAC Address, PHY address, Ethernet Address*
- *Stick to the device/NIC*
- *48 bit, or 6 groups of Hex numbers*
- *Range of each byte: 00 to FF*
- *Special addresses:*
 - *00:00:00:00:00:00 - unknown*
 - *FF:FF:FF:FF:FF:FF - broadcast*
 - *01:00:5E:XX:XX:XX - groupcast (historically only half of these addresses(23 bit))*

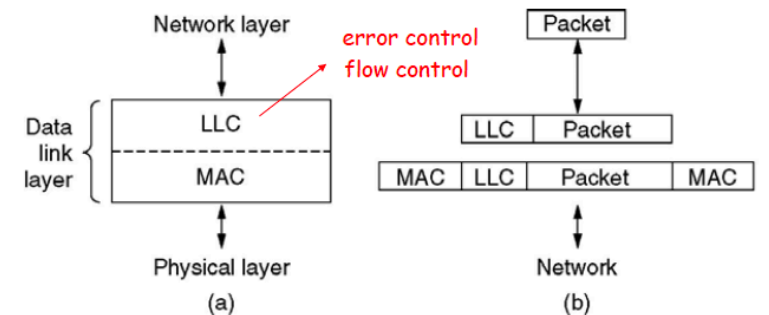
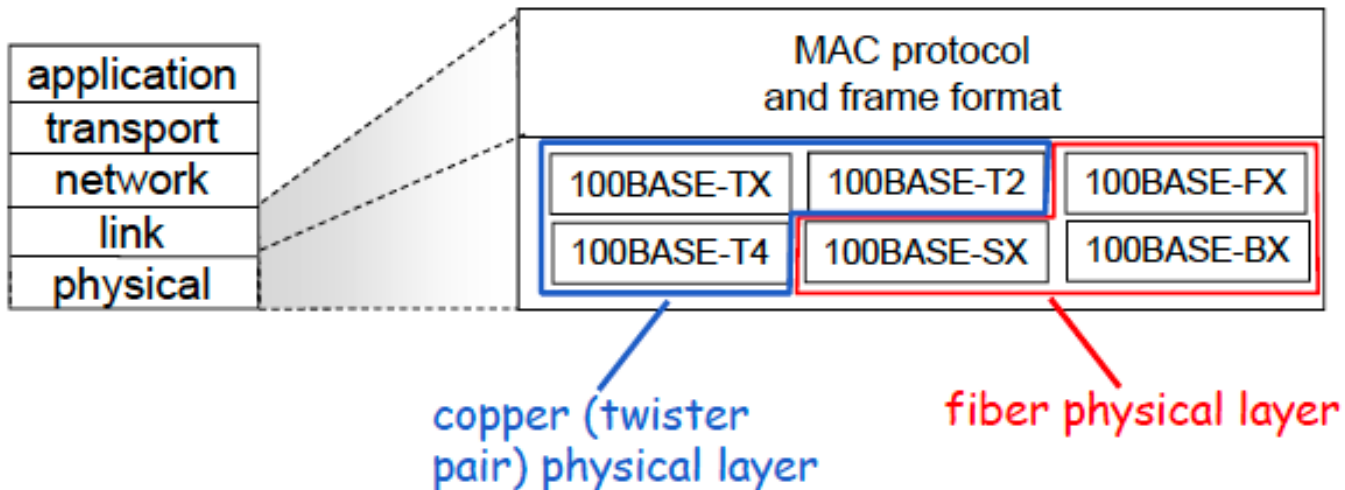
4 Modes of Network Interface Card (NIC)

- **Broadcast:** accept *broadcast* frames (address FF:FF:FF:FF:FF:FF)
- **Multicast:** accept *all groupcast* frames (even if it is not a member of the group)
- **Direct:** accept frames only destined to itself (unicast address)
- **Promiscuous 混杂:** accept *all* frames - that's how Wireshark works
 - NIC of a switch/bridge works in this mode.
- **Note**
 - default mode of a NIC: Broadcast + Direct



802.3 Ethernet Standards: Link and PHY

- **There are many different Ethernet standards**
 - Common MAC protocol and frame control
 - Different speeds: 2 Mbps, 10 Mbps, 100 Mbps, 1 Gbps, 10 Gbps
 - Different PHY media: (**T**wisted pair) cable, **F**iber



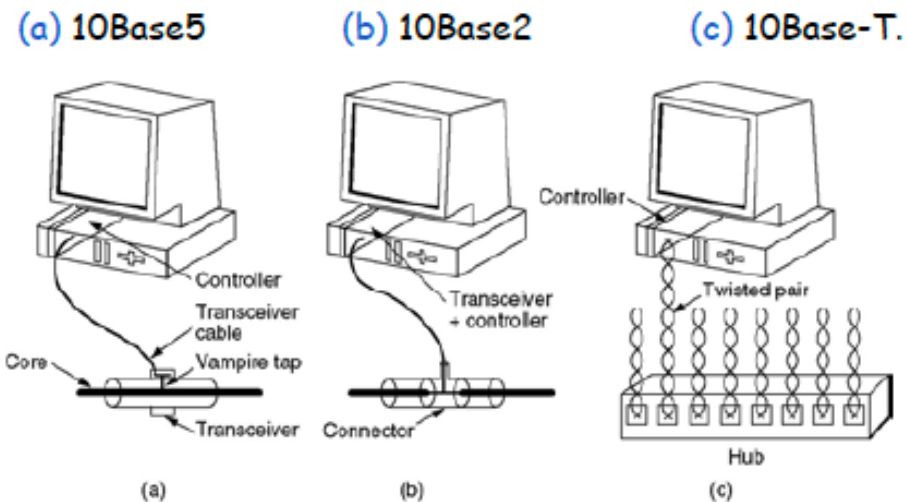
(a) Position of LLC.

(b) Protocol formats.

IEEE 802.2 Logic Link Control

Cables in the Ethernet

- **Old ones**



Name	Cable	Max. seg.	Nodes/seg.	Advantages
10Base5	Thick coax	500 m	100	Original cable; now obsolete
10Base2	Thin coax	185 m	30	No hub needed
10Base-T	Twisted pair	100 m	1024	Cheapest system
10Base-F	Fiber optics	2000 m	1024	Best between buildings

- **Currently in use**

Check your NIC:
ethtool eth5

- **100Base-T: 100Mbps Ethernet**

- 100Base-TX <- 10Base-T5
- 100Base-FX <- 10Base-FL
- Max Seg. 205m

- **1000Base-T: Gigabit Ethernet**

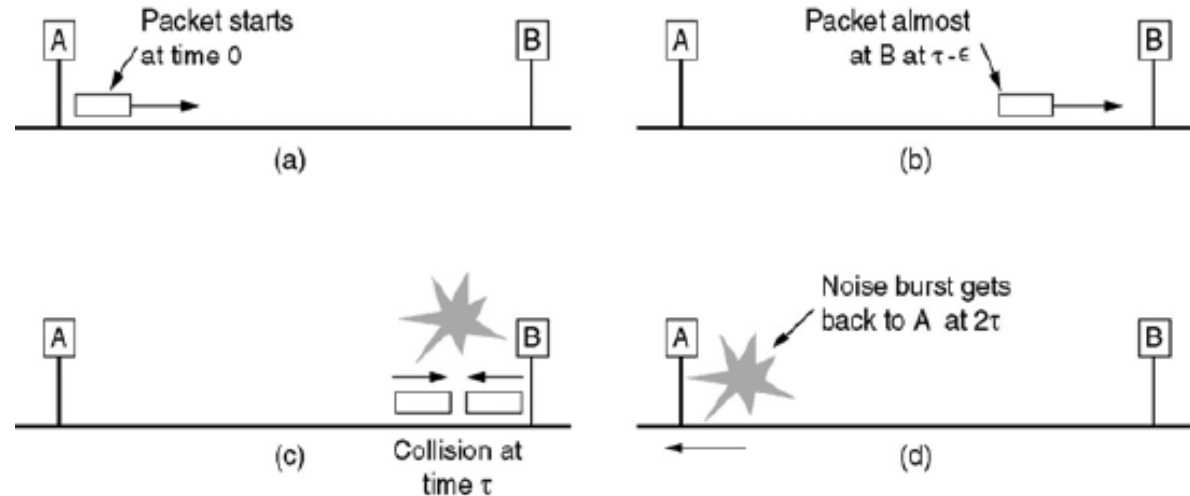
- **2.5GBase-T, 5GBase-T**

- **10GBase-T 万兆以太网**

Name	Cable	Max. segment	Advantages
1000Base-SX	Fiber optics	550 m	Multimode fiber (50, 62.5 microns)
1000Base-LX	Fiber optics	5000 m	Single (10 μ) or multimode (50, 62.5 μ)
1000Base-CX	2 Pairs of STP	25 m	Shielded twisted pair
1000Base-T	4 Pairs of UTP	100 m	Standard category 5 UTP

The Minimum Frame Length

- **The Min. frame length is used to stop transmission w/ collision.**
 - Too short -> *transmission may be completed w/o knowing there was a collision.*
 - So transmission time of one frame $\geq \max RTT \Rightarrow L_{min}/R \geq 2d/c$
 - trans. delay L_{min}/R
 - prop. delay (one hop) d/c
 - E.g. 10Base-5 Ethernet
 - $d=2500m$, $RTT=50\mu s$
 - $R=10\text{ Mbps}$
 - $L_{min} \geq 500\text{ bits} \sim 64\text{ Bytes}$
 - $R \uparrow \rightarrow L_{min} \uparrow$ and $d \downarrow$



Collision detection can take as long as 2τ

Ethernet: unreliable, connectionless services

- **Connectionless: no handshaking**
- **Unreliable: No ACK or NAK (no ARQ)**
 - *Erroneous frames with incorrect CRC will be dropped.*
 - *There may be gaps in the stream of datagrams passed to the network layer, and this gap will be filled if TCP is used on the transport layer.*
 - *Otherwise (UDP is used), the application will see these gaps.*
- **Ethernet's MAC: unslotted CSMA/CD (CSMA/CD w/ modification)**
 - *carrier sensing, collision detection*
 - *but with **random access** and **exponential backoff***

*Works well w/ TCP/IP -
Both IP and Ethernet are
connectionless!*

Ethernet MAC: CSMA/CD (1/2)

1. NIC receives datagrams from the network layer, and creates a frame.
2. (**CSMA**) If NIC senses the channel **idle** (*idle for 96 bit time*), it starts to transmit. If it senses the channel busy, it waits until the channel idle and then transmits.
3. (**CD**) If NIC successfully transmits the entire frame w/o detecting another transmission (i.e. no collision), it is done with the frame.
4. (**CD**) If NIC detects a collision, it aborts and sends a **jam signal** (48 bits long).

NIC: Network Interface Card

bit time: the time to transmit 1 bit, e.g. $0.1 \mu\text{s}$ for 10 Mbps Ethernet.

jam signal: 48 bits long, to make sure all other transmitters are aware of the collision.

Ethernet MAC: CSMA/CD (2/2)

5. (**Exponential Backoff**) After aborting, NIC enters **exponential backoff**, i.e. after the m^{th} collision, NIC chooses a number K uniformly at random from $\{0, 1, 2, \dots, 2^m - 1\}$, waits $K \cdot 512$ bit-time, and returns to Step 2.

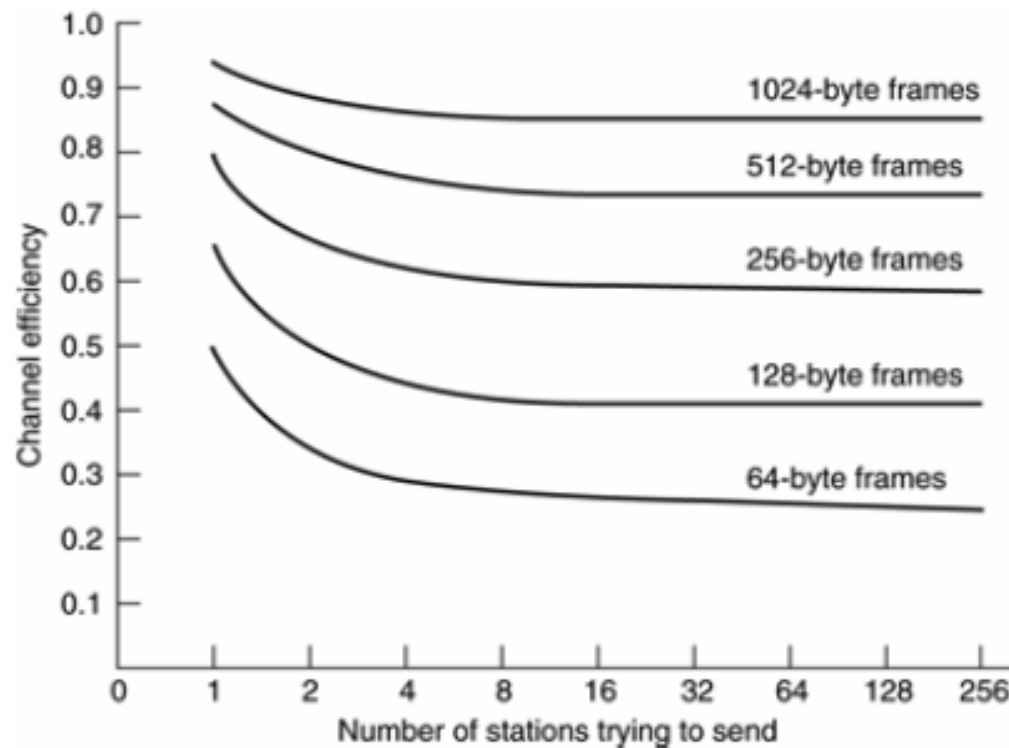
Prob. of
selecting $K=k$?

Exponential Backoff 指数(随机)退避:

- **Goal:** adapt retransmission attempts to estimate the current load.
 - Heavy load $\rightarrow$ waits longer
- **Procedure:** increases the range of K
 - 1st collision: chooses K from $\{0, 1\}$.
 - 2nd collision: chooses K from $\{0, 1, 2, 3\}$.
 - ...
 - 10th collision: chooses K from $\{0, 1, 2, 3, \dots, 1023\}$.
 - 16th collision: reports failure.

Efficiency of the Ethernet MAC

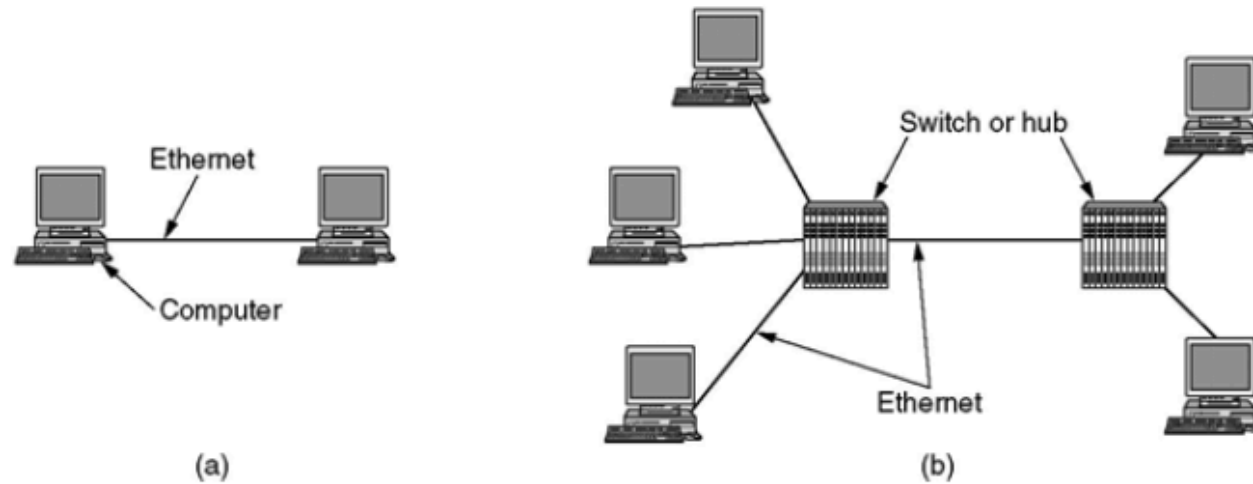
- **About the same as that of CSMA/CD**



Why not infinite frame length to increase efficiency?

Ethernet Topology - Connecting an Ethernet

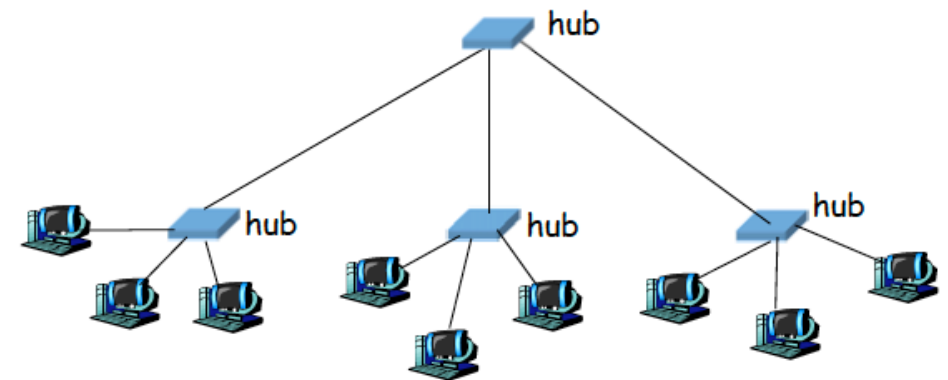
- **Star** topology
- Need a **switch/hub** to connect multiple stations/hosts



(a) A two-station Ethernet. (b) A multistation Ethernet.

Hubs

- **PHY layer “dumb” repeaters (extends max distance)**
 - Bits coming in one link go out *all other links at the same rate*
 - No CSMA/CD at the hub
 - Host NIC is responsible for collision detection.
 - All nodes connected to the same hub can collide with one another.
 - Individual segment collision zones become *one large collision zone*.
 - No frame buffering
 - Can not interconnect 10Base-T or 100Base-T



Switches

- **Link layer devices (“stores-and-forwards” Ethernet frames)**

- *Buffer frames: Store-and-Forward*

- *Examines frame header and forwards frames based on MAC addresses.*

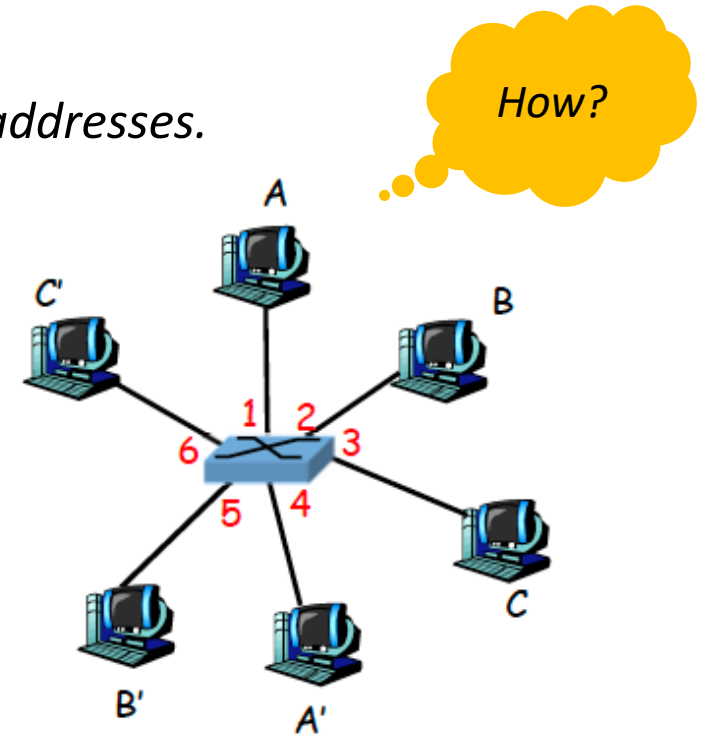
- *CSMA/CD on separate collision zones*

- *When frame is forwards, use CSMA/CD to access segments.*
 - *Ethernet protocol is used on each incoming link, full duplex.*
 - *Each link belongs to its own collision zone.*

- *Transparent to hosts*

- *Switching: A-to-A' and B-to-B' can go simultaneously.*

- *No configuration needed: plug-and-play, self-learning*



*switch with six interfaces
(1,2,3,4,5,6)*

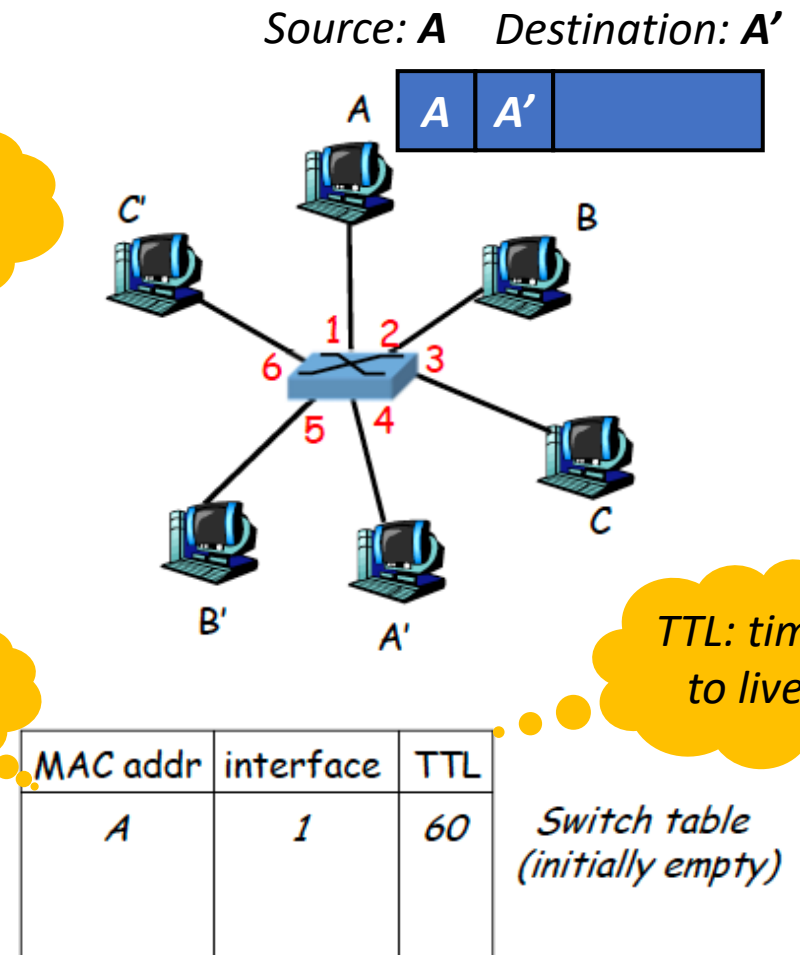
Switching Table and Self-learning

- **Each switch has a switching table.**

- Entry: MAC address of a host, interface to each host, and a time stamp.
- Looks like a routing table, but different.
- Create and maintain: Self-learning
 - When a frame is received, the switch "learns" the location of the sender (incoming LAN segment).
 - It records the sender-interface pair in the switching table.

Routers:
next
chapter

Index
w/ MAC



Switch Filtering/Forwarding

- **When switch *receives* a frame:**

index switch table using MAC dest. address

if entry found for destination

then{

if dest. is on segment from which the frame has arrived

then drop the frame

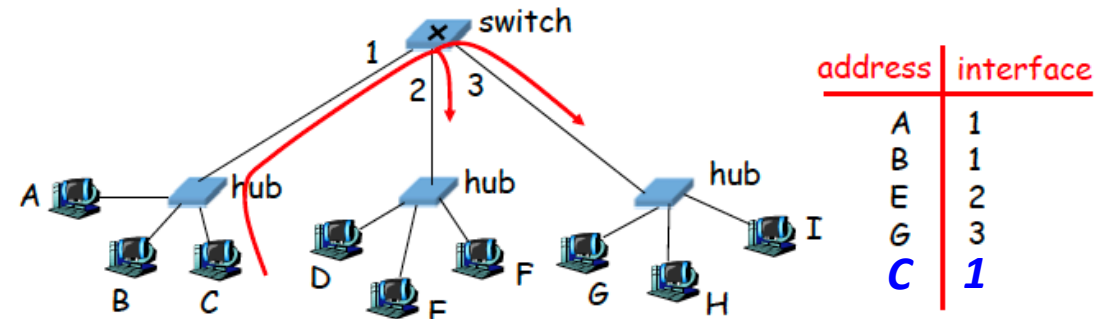
else forward the frame on interface indicated by the switching table

}

else flood

Flood: Forward to all interfaces except the one there the frame has arrived.

Suppose C sends a frame to D



- **Switch receives a frame from C**

- **Writes** in the switching table: Host C is on interface 1.

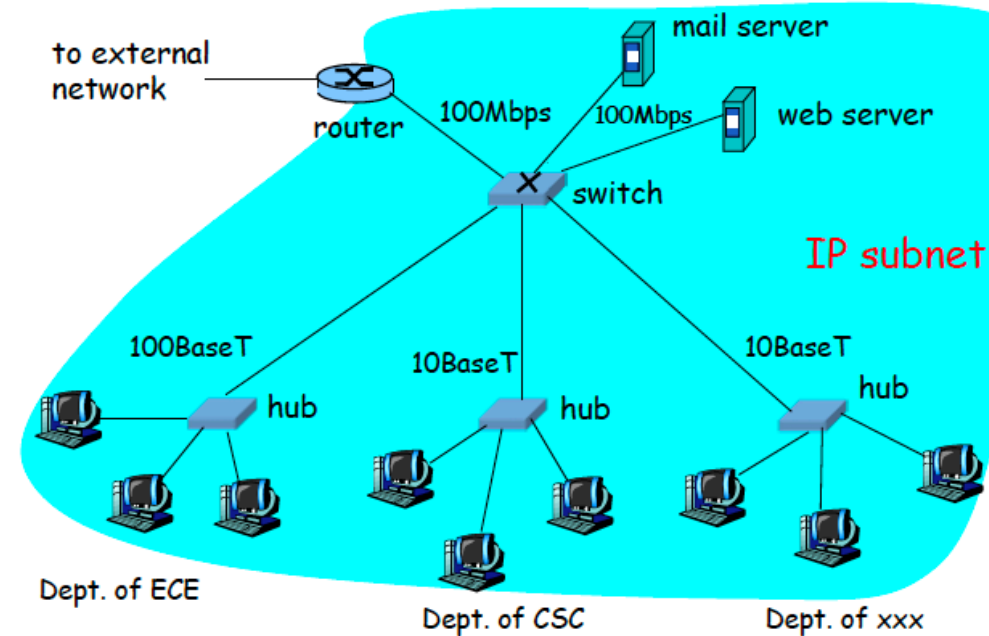
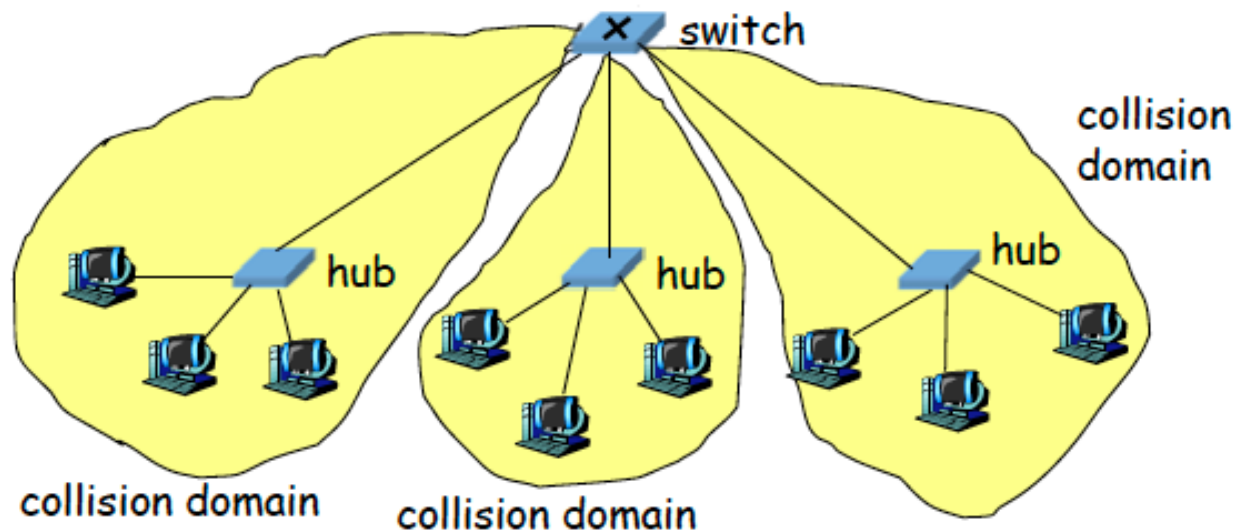
- D is not in the table -> **flood** to interface 2 and interface 3.

When D replies?

Switch breaks subnets into LAN segments

- **Switch filters frames**

- Frames from the same LAN-segment are not forwarded onto other LAN-segments
- LAN-segments -> separate collision domains/zones



E.g. an institutional network

Network Devices

- **Modem**

- PHY
- Analog <-> Digital

- **Hub/Repeater**

- PHY
- Broadcast to all other ports
what it receives from one port
- One contention zone
- Broadcast storm

- **Switch/Bridge**

- Link layer -> use MAC address
- Connect different network segments
-> multiple contention zones
- Filter: local frames stay local
- Bridge may have broadcast storm;
Switch does not (Spanning tree).

- **Router**

- Network layer -> use IP address
- Connect different subnetworks



3.4

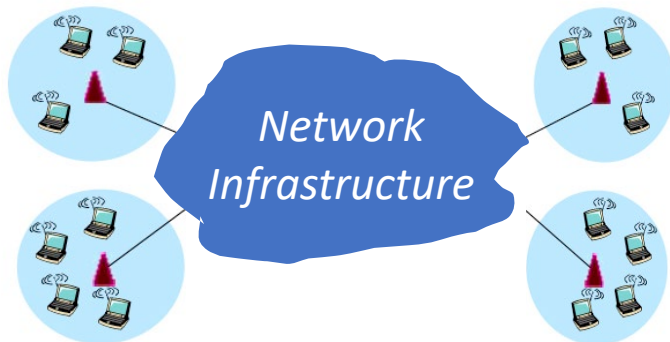
Wireless LAN (WLAN)

LAN in the Wireless Network

- **Wireless network: access network** • **IEEE 802.11 Wireless LAN**

- Base station 基站

- typically connected to wired networks
- relay: sending packets between wired networks and wireless hosts in a certain “region” -> providing access
- E.g. cell towers, 802.11 access point (AP)



- All use **CSMA/CA**

- All have base station and ad-hoc modes

- 802.11n (WiFi 4)

- 2.4-5 GHz unlicensed spectrum bands
- up to 600 Mbps

- 802.11ac (WiFi 5): 5 GHz, 3.4 Gbps

- 802.11ax (WiFi 6): 2.4/5 GHz, 7 Gbps

- 802.11be (WiFi 7): 2.4/5/6 GHz, 46 Gbps

- 802.11bn (WiFi 8): 2.4/5/6/42/71 GHz,
100 Gbps

2026/3/26

Wireless Link Characteristics

- **Key differences**

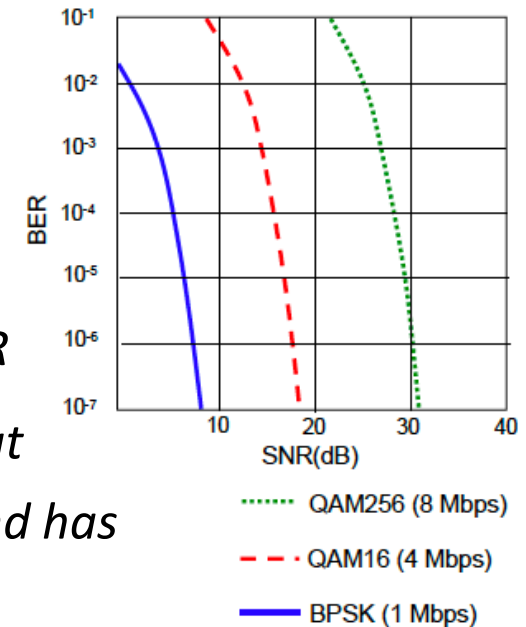
- **Attenuation** 衰减: decreased signal strength in propagation
- **Interferences** 干扰: Frequency bands are shared.
- **Multi-path** 多径效应
propagation: Multiple copies of (reflected) signals superpose.

- **Communication is difficult.**

- **SNR matters!**

- **signal-to-noise ratio**
- Larger SNR -> Easier to extract signal from noise -> good
- **SNR and BER**
 - **Given PHY**: increases power -> increases SNR -> decreases BER
 - **Given SNR**: choose the PHY that meets the BER requirement, and has the highest throughput

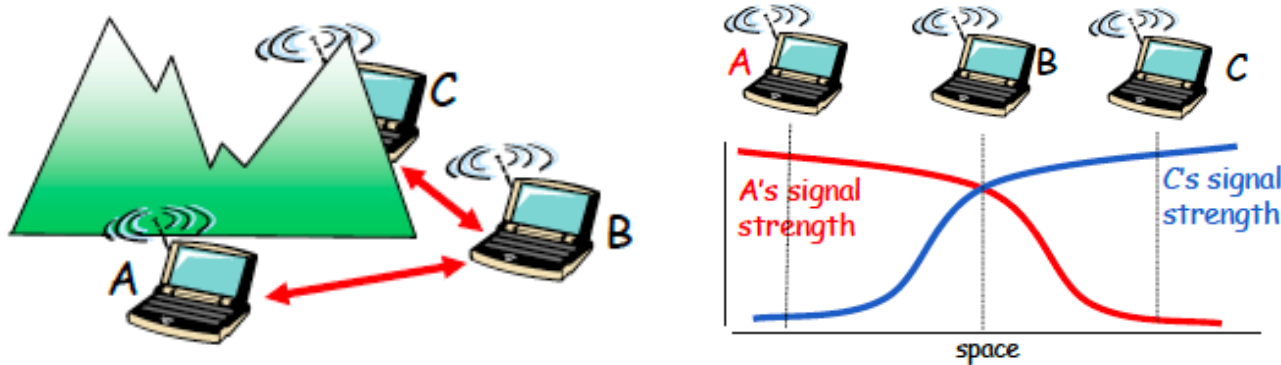
Recall
Shannon
Capacity



Wireless Network Characteristics

- Multiple senders and receivers:

- **Hidden Terminal Problem**



- A and B hear each other
- B and C hear each other
- A and C do not hear each other
- A and C may interfere each other at B

The problem is not collision, but *not knowing* there was a collision.

- CSMA still works, but **no collision detection**.
- Difficult to sense collision due to fading
- Sometimes may not sense collision at all: the hidden terminal problem
- **Goal: Avoiding Collision**
- CSMA/CA - collision avoidance

IEEE 802.11 MAC: CSMA/CA

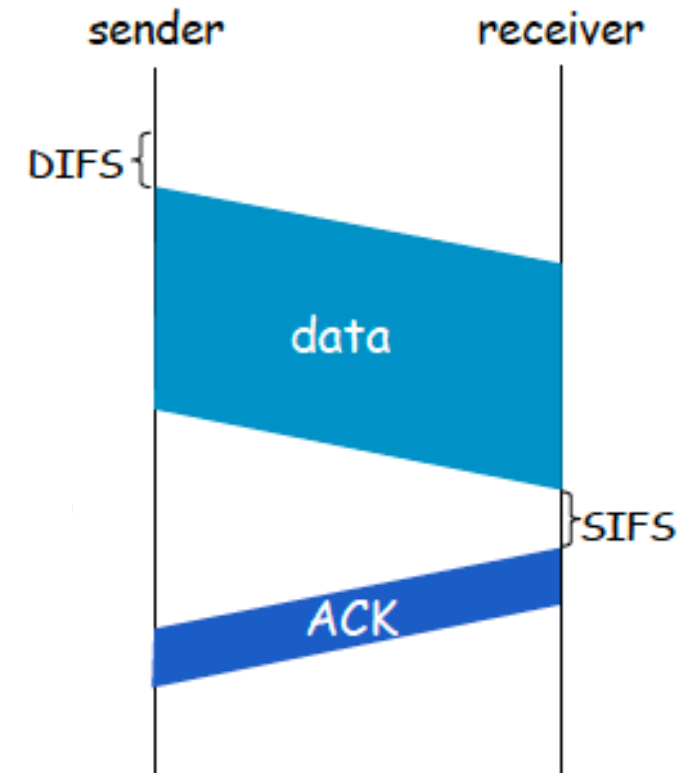
• Sender

1. If senses channel idle for **DIFS**, then transmit the entire frame.
2. Otherwise (busy), then
 - Start **random backoff** (by starting a timer).
 - Timer counts down while channel idle.
 - Timer freezes while channel busy.
 - **Transmit when timer expires.**
 - If ACK is not received, **increase backoff interval**, repeat step 2.

• Receiver

- If the frame is successfully received, replies an ACK after **SIFS**.

Exponentially!



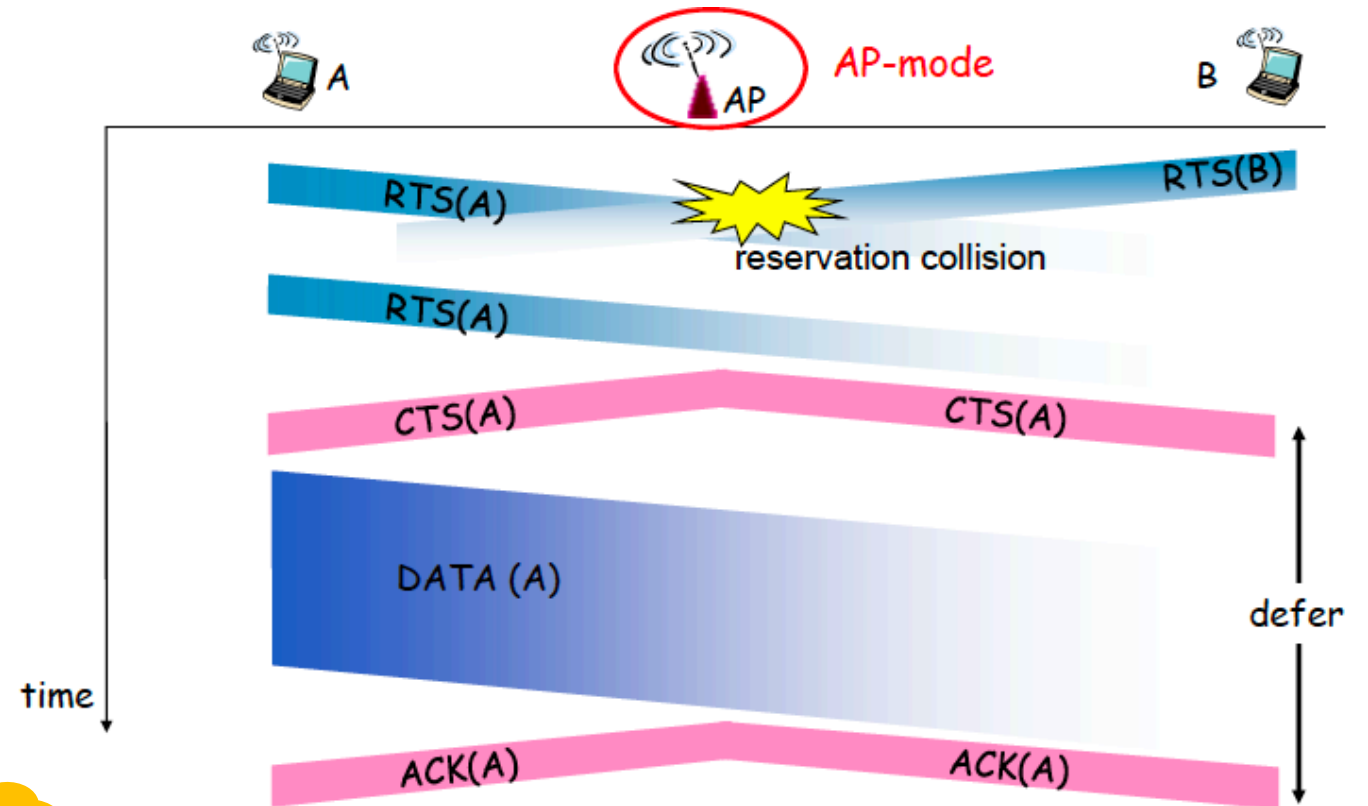
Collision Avoidance: RTS-CTS

- **Idea:** sender “*reserves*” the channel to avoid collision of *long* frames.

- **RTS** (Request-to-send), **CTS** (Clear-to-send)

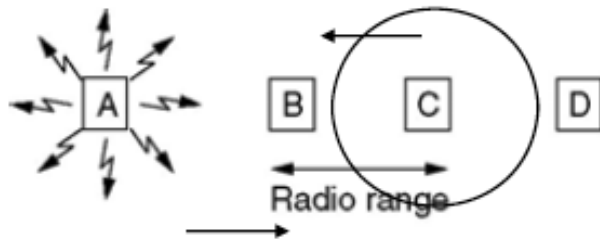
- **Sender:** sends RTS to AP/BS.
 - RTS may collide, but it's short.
- **BS:** broadcasts CTS in response to the RTS.
- **Sender:** transmits the data.
- **Other nodes who hear CTS:** don't transmit.

Ad hoc mode?

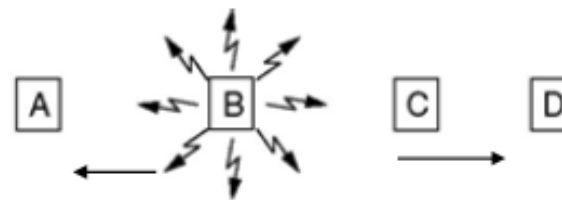


Ad Hoc Mode

- **No AP to coordinate - every host is equal.**
- **Problems: *hidden terminal*, *exposed terminal*.**



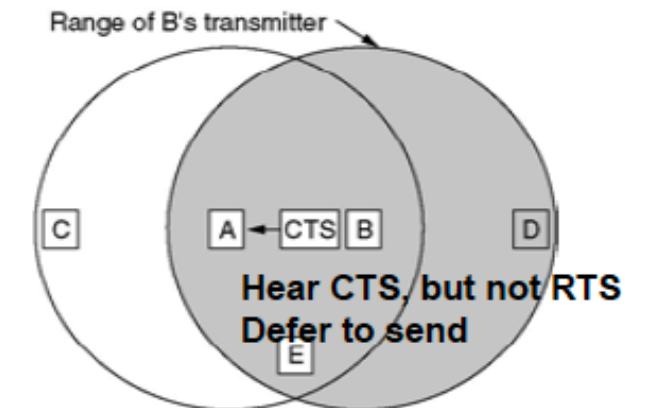
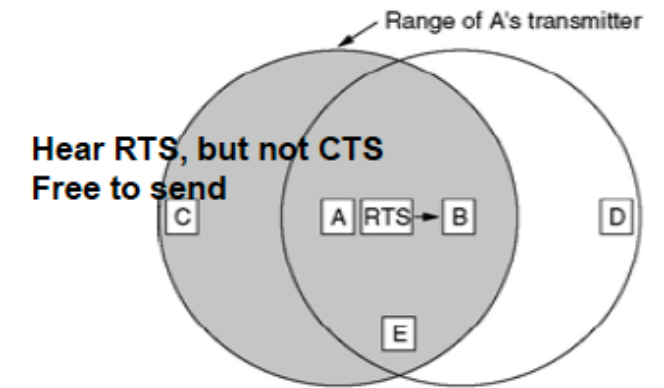
C can not hear A.



*C hears B, and thinks it
can not send to D.*

- **Multiple Access w/ Collision Avoidance:**

- Same as AP mode: A sends RTS. B replies CTS.
- Other nodes: hears CTS -> keep quiet; *hears RTS but not CTS -> OK to transmit.*



Summary

- **Medium Access Control - sublayer of the Link layer**

- *Function: allocate access to a shared channel.*

- *Types:*

- *Fixed allocation: “XDMA”*

- *Random access:*

- **ALOHA**: pure, slotted

- **CSMA**: p-persistent, non-persistent

- **CSMA/CD** -> **Ethernet** (+ Exponential backoff)

- **CSMA/CA** -> **Wireless LAN** (+ACK, RTS/CTS)

- *Taking turns: polling, token-ring*

- *Limited contention: reservation, binary count down*

Performances:

- *Throughput*
- *Access/wait time*
- *Fairness*

- **Network Devices**

- *Hub v.s. Switch v.s. Bridge*

- *Router*

- **MAC Address**

- *Unknown*

- *Unicast*

- *Groupcast*